

Portable Fault Diagnosis for RAG-Based Support Pipelines

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Abstract

Retrieval-Augmented Generation (RAG) pipelines deployed in customer-support settings are susceptible to failures at distinct processing stages: the retriever may surface irrelevant documents, the chunking module may truncate or contaminate the retrieved passage, and the language model may produce an answer that contradicts or ignores the provided context. Localizing the culpable component from a pipeline execution log is non-trivial, yet is essential for systematic maintenance and quality control. This paper presents a portable fault-diagnosis system developed for the SMILES 2026 challenge, which requires trace-level and pipeline-level triage across six fault categories — *healthy*, *retrieval*, *chunking*, *generation*, *out_of_scope*, and *unknown* — under a stringent portability constraint: the system is trained on English open-domain data and evaluated on Russian banking-support data unseen during development. We describe the problem formulation, the label taxonomy, the input trace schema, and a content-based diagnostic methodology that operates without artifact exploitation, yielding a single-entry-point evaluation script (`evaluate.py`) whose outputs are reproducible across independent runs.

Index Terms: Retrieval-Augmented Generation, fault localization, pipeline diagnosis, portability, knowledge-base support systems.

1 Introduction

Retrieval-Augmented Generation [15] has become the dominant architecture for knowledge-intensive question answering in enterprise settings, coupling a neural language model with a document retriever to ground generated responses in authoritative external sources. In customer-support deployments, a RAG assistant processes a client query by searching a knowledge base, extracting text passages, and synthesizing a final answer from the retrieved evidence. When the quality of the assistant’s responses degrades, diagnosing the culpable pipeline component — retriever, chunker, or language model — is a prerequisite for targeted remediation.

The SMILES 2026 challenge, proposed by Alfa Advanced Analytics, formalizes this diagnostic task as a multi-class classification problem. Given an execution trace comprising the original client query, the set of retrieved document chunks, an empty tool-call list, and the final generated answer, a diagnostic system must assign one of six fault labels at the trace level and aggregate these into a pipeline-level verdict with an estimated fault prevalence $p \in [0, 1]$.

The central axis of the challenge is *portability*: the diagnostic system is developed on an open-domain English training corpus, yet evaluated on a closed Russian banking-support corpus. A solution that over-fits the statistical patterns of the training distribution — including language, topic distribution, or artifact features such as pipeline identifiers — will generalize poorly to the

evaluation set. This constraint explicitly penalizes exploitation of non-content features and rewards the development of content-grounded diagnostic signals.

1.1 Problem Statement

Let a trace τ be the tuple (query, retrieved, final_answer, pipeline_id), where retrieved is an ordered list of ranked document chunks. A pipeline Π is the collection of all traces sharing a common pipeline_id.

The diagnostic task is twofold:

- **Trace-level classification.** Assign a label $y_\tau \in \mathcal{Y}_{\text{trace}}$ to every trace τ .
- **Pipeline-level classification.** Assign a systematic fault label $y_\Pi \in \mathcal{Y}_{\text{set}}$ and a prevalence estimate $\hat{p}_\Pi \in [0, 1]$ to every pipeline Π .

The label sets are defined as follows:

$$\begin{aligned}\mathcal{Y}_{\text{trace}} &= \{\text{healthy, retrieval, chunking,} \\ &\quad \text{generation, out_of_scope, unknown}\}, \\ \mathcal{Y}_{\text{set}} &= \mathcal{Y}_{\text{trace}} \setminus \{\text{out_of_scope}\}.\end{aligned}$$

A key invariant is that $y_\tau \neq y_\Pi$ in general: a pipeline labeled healthy at the set level may contain individual traces with non-healthy labels (background noise), while a pipeline with a systematic fault has exactly one dominant fault type and only a fraction p of traces exhibiting that fault.

1.2 Fault Taxonomy

Table 1 summarizes the six fault classes and the diagnostic criterion for each.

Table 1

Fault label taxonomy. Each class corresponds to a distinct failure locus in the RAG pipeline.

Label	Diagnostic criterion
healthy	Query, retrieved chunk, and answer are mutually consistent.
retrieval	Retrieved document is topically irrelevant to the query.
chunking	Document is relevant but the chunk is truncated, fused, or otherwise corrupted.
generation	Chunk is relevant and intact, but the answer contradicts or ignores it.
out_of_scope	Query cannot be resolved via the knowledge base and requires external data or clarification.
unknown	Query text is technically corrupted prior to retrieval, causing loss of semantic content.

1.3 Significance and Contributions

The primary scientific contribution of this work is a methodology for content-based, cross-lingual fault localization in RAG pipelines. Concretely, we:

- **C1:** Formalize the trace-level and pipeline-level diagnostic problem with a precise label taxonomy that disentangles retrieval, chunking, and generation faults.
- **C2:** Propose a cascade of content-grounded diagnostic signals — semantic relevance, chunk coherence, and answer faithfulness — that operate independently of language and domain.
- **C3:** Deliver a single-entry-point, reproducible evaluation script (`evaluate.py`) that runs in an isolated, offline environment against an OpenAI-compatible model endpoint.
- **C4:** Report a planned evaluation on the closed Russian banking corpus under the four scoring criteria defined by the challenge organizers.

2 Background & Literature Review

Retrieval-Augmented Generation. RAG [15] was introduced to address the brittle parametric knowledge of neural language models by conditioning generation on documents retrieved at inference time. Retrievers range from sparse lexical methods (BM25 [19]) to dense bi-encoders such as DPR [13], and hybrid schemes that combine both signals. Downstream benchmarks have evolved from single-hop factoid QA (Natural Questions [14], TriviaQA [12]) to multi-hop settings that require reasoning across multiple retrieved passages (HotpotQA [23], MuSiQue [22]). Enterprise deployments of RAG, particularly in customer-support contexts, introduce additional complexity: the knowledge base is a curated product catalog or policy corpus, queries are often fragmentary, and a wrong or evasive answer carries reputational risk.

Failure modes in RAG pipelines. Prior work identifies three loci at which RAG quality can degrade [1]. *Retrieval failure* occurs when the top-ranked passages do not address the query, forcing the generator to produce a refusal or hallucination without grounding evidence. *Context-processing failure* encompasses both chunking artifacts (truncated or concatenated passages [7]) and over-long context dilution. *Generation failure* covers faithfulness violations, where the model contradicts or selectively ignores the provided context, a phenomenon studied under the rubric of hallucination [16] and retrieval-generation misalignment [21]. Our taxonomy refines this three-way decomposition by adding *out_of_scope* (queries principally unresolvable via the knowledge base) and *unknown* (technically corrupted input signals), both of which are practically important in deployed systems.

Faithfulness and relevance evaluation. The automated evaluation of answer faithfulness to source documents underpins both RAG evaluation frameworks [8] and fact-verification research [11]. Natural language inference (NLI) [2] models trained to classify entailment, neutrality, and contradiction between a premise and a hypothesis provide a model-internal proxy for faithfulness. Cross-lingual NLI models [5, 10] extend this capability to non-English languages, making them attractive for portability-critical applications. Semantic similarity methods based on dense encoders [18] offer a complementary signal for retrieval relevance assessment.

Cross-lingual and domain transfer. A central challenge in

the SMILES 2026 task is that the diagnostic system must generalize from English open-domain data to Russian banking data without supervision on the target distribution. This setting is closely related to zero-shot cross-lingual transfer [17], where a model fine-tuned on one language is applied to another. Multilingual pre-trained encoders such as mBERT [6] and XLM-R [4] have demonstrated strong zero-shot transfer for classification tasks, particularly when the classification signal is rooted in semantic content rather than surface lexical patterns.

Automated pipeline monitoring and root-cause analysis.

The broader context for this work is the automated quality monitoring of machine learning pipelines in production. Data-centric methods [9] detect distributional shift in model inputs and outputs; slice-based evaluation [3] identifies subpopulations where quality systematically degrades. Our fault-diagnosis task is a specialized instance of this paradigm, where the “slices” correspond to the structural components of a RAG pipeline and the diagnostic signal is derived from the semantic coherence of the execution trace rather than from aggregate statistics.

Gap. Existing RAG evaluation frameworks (e.g., RAGAS [8], ARES [20]) primarily assess the *quality* of generated answers rather than performing fault localization at the component level. They do not address the portability requirement, nor do they produce structured, machine-readable verdicts suitable for systematic pipeline triage. The SMILES 2026 challenge fills this gap by casting fault localization as a labeled classification problem evaluated on held-out, cross-lingual data.

3 Methods

Overview. The diagnostic system operates as a three-stage cascade over each trace $\tau = (\text{query}, \text{retrieved}, \text{final_answer})$. Stage 1 performs pre-screening to detect degenerate inputs (*unknown* and *out_of_scope*). Stage 2 assesses retrieval relevance. Stage 3 jointly evaluates chunk integrity and answer faithfulness. Pipeline-level verdicts are derived by aggregating trace-level labels within each *pipeline_id* group. The diagnostic criteria are grounded exclusively in the textual content of the trace fields; no artifact features (file order, pipeline letter, chunk-count statistics, or raw length distributions) are used.

Stage 1 — Input screening. Two conditions are tested prior to retrieval analysis:

Unknown detection. The query field is examined for technical corruption that causes irreversible semantic loss: encoding-garbled characters, mid-word truncation that eliminates the topical kernel of the query, or a combination of partial transliteration and abbreviation such that the query intent cannot be recovered. Surface imperfections that preserve intent (typos, colloquial spelling, code-switching) do not trigger this class. Formally, a query is classified as *unknown* if a language model judge, given both the raw query string and the retrieved passages, reports that the query topic cannot be inferred. This is the only class tested solely from the query field; all others require joint analysis of the full trace.

Out-of-scope detection. A query is classified as *out_of_scope* if it is semantically well-formed but inherently unresolvable via a static knowledge base: it requires personalized client data (account balances, transaction history), an action in an external system, or is underspecified to the point where no knowledge-base article could constitute a correct answer. The signal is that no pos-

sible retrieved passage — regardless of retrieval quality — would suffice to produce a grounded, correct response.

Stage 2 — Retrieval relevance assessment. For traces that pass Stage 1, the topical alignment between the query and the top-ranked retrieved chunk is evaluated via a multilingual semantic similarity model (cross-lingual sentence encoder). A low similarity score, combined with a final answer that is a refusal, a generic disclaimer, or topically unrelated to the query, yields the label retrieval. The boundary with `out_of_scope` is respected: if the query is principally unresolvable, Stage 1 pre-empts this stage.

Stage 3a — Chunk integrity assessment (chunking). For traces where retrieval is judged relevant, the retrieved chunk is inspected for structural integrity. Two chunking defects are targeted:

- *Truncation*: the chunk terminates mid-sentence or mid-clause, leaving a key fact absent (e.g., a sentence that begins “The daily withdrawal limit is” but supplies no figure).
- *Fusion*: the chunk concatenates semantically disjoint passages from different sections of the source document, injecting noise into the context.

Detection relies on coherence analysis: a language model judge assesses whether the chunk is self-contained and internally consistent with respect to the query. A chunk from the correct source document (`doc_id` semantically consistent with the query) that nonetheless fails coherence is labeled chunking.

Stage 3b — Answer faithfulness assessment (generation). If the chunk is intact and the retrieval is relevant, answer faithfulness is evaluated via a Natural Language Inference (NLI) classifier [11] treating the chunk as the premise and the final answer as the hypothesis. A contradiction verdict, or a finding that the answer introduces facts not entailed by the chunk (hallucination), yields the label generation. Traces where the chunk is sufficient, the answer is consistent with it, and no upstream fault is detected are labeled healthy.

Pipeline-level aggregation. Given a set of trace-level labels $\{y_\tau\}_{\tau \in \Pi}$ for pipeline Π , the pipeline-level fault component y_Π is determined as follows. Let $f_c = |\{\tau : y_\tau = c\}|/|\Pi|$ be the fraction of traces labeled with class c for fault classes $c \in \{\text{retrieval, chunking, generation, unknown}\}$. The pipeline is labeled with the dominant fault class $c^* = \arg \max_c f_c$ if f_{c^*} exceeds a background noise threshold θ ; otherwise the pipeline is labeled healthy. The prevalence estimate is $\hat{p}_\Pi = f_{c^*}$ for faulted pipelines and $\hat{p}_\Pi = 0$ for healthy ones. The threshold θ is calibrated on the training corpus to maximize the pipeline-level F-measure.

Reproducibility and constraints. The complete system is encapsulated in a single-entry-point script:

```
python evaluate.py <data_dir> -out report.json
```

The script reads all `pipeline_*.jsonl` files from `<data_dir>`, invokes the language model judge via a single OpenAI-compatible endpoint (address and model name supplied at runtime), and writes the structured output `report.json`. No internet access beyond the model endpoint is required. Diagnosis is based entirely on trace content; the number of retrieved chunks per trace is held constant by the organizers and is not used as a discriminative feature.

4 Results

The evaluation protocol applies the diagnostic system to the closed Russian banking-support corpus held by the challenge organizers, comprising approximately nine pipelines and up to 1 000 traces in total. Results are reported under the four weighted scoring criteria defined in the challenge specification; cells in the tables below will be populated upon the official evaluation run.

Table 2

Planned trace-level evaluation results. Macro-averaged F-measure (class-averaged) over the held-out subset of traces. Column headers correspond to per-class F1 scores; the rightmost column reports the macro average.

System	hlthy	retr	chnk	gen	oos	unk
Majority baseline	—	—	—	—	—	—
Stage-1 only	—	—	—	—	—	—
Full cascade	—	—	—	—	—	—

Table 3

Planned pipeline-level evaluation results. *B*: accuracy of pipeline component identification (including unknown). *C*: mean absolute error of prevalence estimate $\hat{p}$. *D*: recall on healthy pipelines (false-alarm rate).

System	<i>B</i> (accuracy)	<i>C</i> (MAE $\hat{p}$)	<i>D</i> (healthy recall)
Majority baseline	—	—	—
Stage-1 only	—	—	—
Full cascade	—	—	—

The completed matrices directly address the design of the cascade. Gains in the retrieval and chunking columns of Table 2 over the majority baseline validate that content-grounded semantic signals (Stages 2 and 3a) transfer to the Russian banking domain. Performance on generation (Stage 3b) reflects the cross-lingual transferability of the NLI-based faithfulness classifier. The unknown column is a direct measure of cross-domain generalization: this class is absent from the training corpus, so any correct identification on the evaluation set constitutes zero-shot transfer. The pipeline-level metric *D* quantifies the practical cost of false alarms: misclassifying a healthy pipeline as faulty may trigger unnecessary engineering interventions.

Reproducibility. The organizers verify reproducibility by running `evaluate.py` independently on a small trace subset; the system is required to produce identical `report.json` outputs across multiple runs. To satisfy this requirement, all stochastic components (language model sampling temperature set to zero, random seeds fixed) are deterministically configured. Computational efficiency is tracked as a secondary metric: at equal accuracy, the solution with the fewest language model invocations prevails, incentivizing the use of heuristic pre-filters in Stages 1 and 2 to avoid unnecessary model calls.

5 Conclusion

This paper presents a portable, content-grounded fault-diagnosis system for RAG-based customer-support pipelines, developed in response to the SMILES 2026 challenge. The system decomposes the diagnostic task into a cascade of three analytically motivated stages — input screening, retrieval relevance assessment, and joint chunk-integrity plus answer-faithfulness evaluation — each targeting a distinct locus of failure in the RAG architecture. Pipeline-

level verdicts are derived by thresholded aggregation of trace-level labels, yielding both a component identifier and a quantitative prevalence estimate $\hat{p}$.

The central design constraint is portability: all diagnostic signals are grounded in the semantic content of the trace (query intent, document relevance, chunk coherence, answer consistency), and no artifact features that are specific to the training distribution are exploited. This design choice is motivated by the explicit cross-lingual, cross-domain evaluation protocol of the challenge, in which solutions that over-fit the English open-domain training corpus are penalized on the Russian banking evaluation corpus. Multilingual pre-trained models provide the backbone for cross-lingual transfer of semantic similarity and NLI-based faithfulness scoring.

Limitations. The cascade architecture sequentially applies threshold decisions, so an error at an early stage propagates forward. The noise threshold θ for pipeline-level triage is calibrated on the training corpus and may not transfer perfectly to the evaluation distribution, particularly for the unknown class, which is absent from training by design. Additionally, the system’s reliance on a single external model endpoint introduces a dependency on the latency and availability of that endpoint during the evaluation run.

Future Work. Several directions merit further investigation. First, end-to-end training of a multilingual diagnostic classifier on the full trace tuple, rather than sequential heuristic stages, may improve consistency. Second, the prevalence estimator $\hat{p}$ could be replaced by a calibrated probabilistic model that accounts for the non-uniform background noise rate across pipelines. Third, the unknown class presents an opportunity for unsupervised anomaly detection, bypassing the need for labeled examples entirely.

Ethical considerations. The system analyzes customer-support interaction logs, which may contain personal or sensitive information. In operational deployment, all trace data should be processed in accordance with applicable data-protection regulations; the diagnostic outputs should be used to improve system quality rather than to profile individual users.

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